

Quiz (Carolina Reaper)

- Q1.** Simplify $\sqrt{16x^2}$ when shipping x crates of goods.
(A) $4x$
(B) $4\sqrt{x}$
(C) $\sqrt{4x}$
(D) $2\sqrt{2x}$
(E) $2x$
- Q2.** Find the value of $\sqrt{18} + \sqrt{8}$ in a traffic flow model.
(A) $\sqrt{26}$
(B) $3\sqrt{2} + 2\sqrt{2}$
(C) $5\sqrt{2}$
(D) 7
(E) $3 + 2\sqrt{2}$
- Q3.** Simplify $\sqrt{50} \cdot \sqrt{2}$ for a logistics delivery system.
(A) 10
(B) $5\sqrt{2}$
(C) $10\sqrt{2}$
(D) $\sqrt{100}$
(E) $\sqrt{25}$
- Q4.** Given $\sqrt{12} = \sqrt{4 \cdot 3}$, what is $\sqrt{12} \cdot \sqrt{27}$ in transportation costs?
(A) $6\sqrt{3} \cdot 3\sqrt{3}$
(B) 18
(C) $6 \cdot 3\sqrt{3}$
(D) $2 \cdot 3\sqrt{3}$
(E) $18\sqrt{3}$
- Q5.** A delivery truck travels x miles in $\sqrt{x}$ hours. How many miles does it travel in $\sqrt{16}$ hours?
(A) 4 miles
(B) 16 miles
(C) $\sqrt{16}$ miles
(D) $\sqrt{4}$ miles
(E) x miles
- Q6.** What is $\frac{\sqrt{20}}{\sqrt{5}}$ in a shipping container's volume calculation?
(A) $\sqrt{4}$
(B) 2
(C) $\sqrt{20}$
(D) $\sqrt{5}$
(E) 4
- Q7.** Simplify $\frac{\sqrt{96}}{\sqrt{6}}$ for a logistics routing system.
(A) $\sqrt{16}$
(B) 4
(C) $\sqrt{4}$
(D) $\sqrt{24}$
(E) $2\sqrt{4}$
- Q8.** Find the value of $\sqrt{24} + \sqrt{9}$ in a traffic management system.
(A) $\sqrt{33}$
(B) $2\sqrt{6} + 3$
(C) $5 + 2\sqrt{6}$
(D) $\sqrt{24} + 3$
(E) 7

Open Ended Questions (FRQ)

- Q9.** A truck travels from City A to City B in $\sqrt{x}$ hours, and from City B to City C in $\sqrt{2x}$ hours. What is the total time traveled in terms of x ?
- Q10.** A shipping company has two types of containers: one with a volume of $\sqrt{128}$ cubic meters and another with a volume of $\sqrt{32}$ cubic meters. What is the total volume of 5 containers of the first type and 3 containers of the second type?

Chef's Tips

- Tip 1: Emphasize the importance of simplifying radicals before performing operations.
- Tip 2: Encourage students to check their calculations for each question.
- Tip 3: Remind students to use the correct mathematical notation and to show their work for free response questions.